TEXORA · LECTURE SHEET

Electricity

Coulomb's law and Gauss's law, the electric dipole, capacitors with and without dielectrics, charging and discharging, Ohm's and Kirchhoff's laws with the Wheatstone bridge, L–R transients, and A.C. through R, L and C — series and parallel resonance.

PHY-1203 · PHYSICS-II · TOPIC 1 · LEVEL 1 TERM 2

1. Coulomb's law and the electric field

Between two point charges in vacuum,

$$F = (1/4\pi\varepsilon_0) \cdot q_1 q_2/r^2, \quad 1/4\pi\varepsilon_0 = 9 \times 10^9 \text{ SI}, \ \varepsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

The **electric field** E is force per unit positive test charge; for a point charge $E = (1/4\pi\varepsilon_0)Q/r^2$, radially out (for $Q > 0$). Fields obey superposition — add vectorially. The **electric flux** through a surface is $\Phi = \int E \cdot dA$: the field's "flow" through the surface.

WORKED EXAMPLES

$q_1 = 2$ μC, $q_2 = 3$ μC, r = 10 cm: $F = 9 \times 10^9 \times (6 \times 10^{-12})/(0.10)^2 =$ **5.4 N** (repulsive).
E at 20 cm from Q = 5 μC: $E = 9 \times 10^9 \times 5 \times 10^{-6}/0.04 =$ **1.13×10^6 N C^{-1}**.

2. Gauss's law and its applications

Gauss's law: the total flux out of any closed surface equals the enclosed charge over ε_0:

$$\oint E \cdot dA = Q_enc/\varepsilon_0$$

Exploiting symmetry turns it into the fastest field-finder in physics:

DISTRIBUTION	E (RESULT)
Point charge / outside a charged sphere	$Q/(4\pi\varepsilon_0 r^2)$
Inside a uniformly charged solid sphere (radius R)	$Qr/(4\pi\varepsilon_0 R^3)$ — grows linearly; zero at centre
Infinite line charge λ	$\lambda/(2\pi\varepsilon_0 r)$, radially out

Infinite charged sheet σ	$\sigma/(2\varepsilon_0)$, independent of distance
Between oppositely charged parallel sheets	σ/ε_0 between; zero outside

3. The electric dipole

A dipole is two equal opposite charges ±q separated by 2a; its **dipole moment** p = 2aq points from −q to +q. On the axis at distance r ($r \gg a$): $E = 2p/(4\pi\varepsilon_0 r^3)$; on the perpendicular bisector (equatorial line): $E = p/(4\pi\varepsilon_0 r^3)$, opposite to p. In an external field E: torque $\tau = pE\sin\theta$ (aligning p with E), potential energy $U = -pE\cos\theta$. The $1/r^3$ fall-off — faster than a charge's $1/r^2$ — is the dipole's signature: from far away the two charges nearly cancel.

4. Capacitors

Capacitance C = Q/V. Parallel plate: $C = \varepsilon_0 A/d$. Combinations: **series** $1/C = \Sigma 1/C_i$ (same charge on each), **parallel** $C = \Sigma C_i$ (same voltage). Energy stored $U = \tfrac{1}{2}CV^2 = Q^2/(2C)$ — the battery does twice this work in charging through a resistor; half is stored, half dissipated as heat (always, whatever R is). With a **dielectric** of constant K filling the gap, C → KC: the dielectric polarises, its bound surface charge cancels part of the plate charge's field, so more free charge fits at the same voltage. If the battery stays connected, U rises K-fold; if isolated, Q is fixed and U falls K-fold — the difference is work done by/against the battery and the field's pull on the slab.

> **WORKED EXAMPLES**
>
> Plates A = 100 cm², d = 1 mm: $C = 8.854 \times 10^{-12} \times (0.010/0.001)$ = **88.5 pF**.
>
> 2 µF and 3 µF: series → (2×3)/(2+3) = **1.2 µF**; parallel → **5 µF**.
>
> 10 µF charged to 100 V: $U = \tfrac{1}{2} \times 10^{-5} \times 10^4$ = **0.05 J**.

5. Charging and discharging; Ohm's law

Through a resistor R, a capacitor charges as $q = Q_0(1 - e^{-t/RC})$ and discharges as $q = Q_0 e^{-t/RC}$; the **time constant** $\tau = RC$ sets the pace (63% in one τ; 99% needs 4.6τ). **Ohm's law**: V = IR for ohmic conductors, from the microscopic picture $J = \sigma E$ with σ the conductivity; resistance $R = \rho L/A$ grows with length, shrinks with area, and for metals ρ rises with temperature.

6. Kirchhoff's laws and the Wheatstone bridge

Junction rule: ΣI = 0 (charge conservation). **Loop rule:** ΣV = 0 round any closed loop (energy conservation). Together they solve any network. The **Wheatstone bridge** — four arms P, Q, R, S in a

diamond with a galvanometer across the middle — balances (no galvanometer current) when

$$P/Q = R/S$$

the laboratory's precision resistance comparator (and the theory behind the meter bridge and P.O. box of the Sessional course).

WORKED EXAMPLE

P = 10 Ω, Q = 20 Ω, R = 15 Ω at balance → S = QR/P = 20 × 15/10 = **30 Ω**.

7. Growth and decay in the L–R circuit

An inductor opposes change: with a battery E connected, the current grows as i = (E/R)(1 − e^{−Rt/L}), time constant τ = L/R; disconnected, it decays as i = i_0e^{−Rt/L}. The missing energy bookkeeping is the inductor's stored energy ½Li², returned as the spark when a switch is opened fast. (Numerical feel: L = 0.5 H, R = 10 Ω → τ = 0.05 s — fifty milliseconds for the current to establish.)

8. A.C. through R, L, C — and resonance

ELEMENT	REACTANCE	PHASE OF CURRENT VS VOLTAGE
Resistor R	R	In phase
Inductor L	X_L = ωL	Lags by 90°
Capacitor C	X_C = 1/(ωC)	Leads by 90°

Series RLC: impedance Z = √(R² + (X_L − X_C)²), current I = V/Z, power factor cosφ = R/Z, average power = V I cosφ (only R consumes power). **Series resonance** at ω_0 = 1/√(LC): X_L = X_C, Z = R minimum, current maximum, power factor 1 — the radio tuner's principle; sharpness measured by the Q factor ω_0L/R.

Parallel resonance (L with its resistance, in parallel with C — the tank circuit): resonance near ω_0 ≈ 1/√(LC) (exactly, (1/L)√(L/C − R²)·… for the practical coil, f_0 = (1/2π)√(1/(LC) − R²/L²)); the impedance is *maximum* and the line current minimum — the rejector circuit, dual of the series acceptor.

WORKED EXAMPLES

L = 0.1 H, C = 10 μF: f_0 = 1/(2π√(LC)) = 1/(2π × 10^{-3}) = **159 Hz**.

Series: R = 30 Ω, X_L = 80 Ω, X_C = 40 Ω, V = 230 V: Z = √(30² + 40²) = 50 Ω; I = 230/50 = **4.6 A**; power factor 30/50 = 0.6, lagging (inductive, since X_L > X_C).

9. Key points

- Gauss’s law + symmetry = fields of spheres, lines and sheets without integration pain.
- Dipole fields fall as $1/r^3$; torque $pE \sin\theta$, energy $-pE\cos\theta$.
- Series capacitors share charge; parallel share voltage. Dielectric multiplies C by K.
- τ_{RC} and τ_{LR} time every transient in this topic; 99% ≈ 4.6τ.
- Wheatstone balance P/Q = R/S is Kirchhoff’s two rules in one ratio.
- Resonance (series): Z minimum, I maximum at $\omega_0 = 1/\sqrt{(LC)}$; parallel: Z maximum, line current minimum.

10. Practice

Answer in your own words; brief model answers follow.

1. Find the force between +4 μC and −2 μC placed 30 cm apart in air, and state its nature.
2. A charge Q sits at the centre of a cube. What is the flux through one face? Through a face if Q moves to a corner of the cube (state qualitatively)?
3. Three capacitors 2, 3 and 6 μF are connected in parallel across 100 V. Find the total charge drawn and the energy stored.
4. A 1 μF capacitor charges through 1 MΩ. How long until it is 99% charged, if the supply would charge it fully to Q_0?
5. In a Wheatstone bridge P = 4 Ω, Q = 6 Ω and R = 9 Ω at balance. Find S.
6. A series circuit has R = 40 Ω, X_L = 90 Ω, X_C = 60 Ω on a 200 V supply. Find the current and the power consumed.

Answer key

1. $F = 9 \times 10^9 \times (8 \times 10^{-12})/(0.30)^2 = 0.072/0.09 = 0.8$ N, attractive (opposite signs).
2. Total flux = Q/ε_0; by symmetry each of the 6 faces carries $Q/(6\varepsilon_0)$. At a corner, the flux is no longer shared equally among the faces (the three faces meeting at the corner get none by symmetry of the solid angle argument — precisely, only 1/8 of the charge’s flux enters the cube at all, since 8 cubes meet at that corner: flux into the cube = $Q/(8\varepsilon_0)$, shared by the three far faces).
3. C_total = 11 μF. $Q = CV = 11 \times 10^{-6} \times 100 = 1.1 \times 10^{-3}$ C. $U = \tfrac{1}{2}CV^2 = \tfrac{1}{2} \times 11 \times 10^{-6} \times 10^4 = 0.055$ J.
4. $\tau = RC = 10^6 \times 10^{-6} = 1$ s. 99% needs $1 - e^{\wedge\{-t/\tau\}} = 0.99 \rightarrow t = \tau \ln 100 = 4.6$ s.
5. $S = QR/P = 6 \times 9/4 = 13.5$ Ω.

6. $Z = \sqrt{(40^2 + (90 - 60)^2)} = \sqrt{(1600 + 900)} = 50\ \Omega$. $I = 200/50 = 4$ A. Power $= I^2R = 16 \times 40 = 640$ W (equivalently $VI\cos\varphi$ with $\cos\varphi = 0.8$: $200 \times 4 \times 0.8 = 640$ W ✓).
